Statistical Machine Learning (CSE342/542/ECE356/556) Assignment 1 – MNIST Classification using MLE and Discriminant Analysis

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1. Methodology

1.1 Data Preprocessing

Following the assignment instructions and standard protocols for the MNIST dataset:

- **Data Split:** The standard MNIST partition using the first 60,000 images as the Training Pool and the last 10,000 images as the Test Pool.
- **Sampling:** Filtered for digit classes 0, 1, and 2. From the Training and Testing Pool, 100 images per class were randomly sampled.
- **Feature Extraction:** The raw 28×28 images were converted into feature vectors by stacking columns. This resulted in a $d = 784$ dimensional feature vector per sample.
- **Normalization:** Pixel values were normalized to the range $[0, 1]$ by dividing by 255.0.

1.2 Maximum Likelihood Estimation (MLE)

Given the data for each class ω_c follows a Multivariate Gaussian distribution:

$$p(x|\omega_c) \sim \mathcal{N}(\mu_c, \Sigma_c)$$

Using the training set ($N_c = 100$ per class), the MLE estimates were computed using the equations:

Mean:

$$\hat{\mu}_c = \frac{1}{N} \sum_{i=1}^{N} x_i$$

Covariance:

$$\hat{\Sigma}_c = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{\mu}_c)(x_i - \hat{\mu}_c)^T$$

Regularization: Since the number of samples ($N = 100$) is significantly smaller than the feature dimension ($d = 784$), the sample covariance matrix is singular (rank-deficient). For invertibility, a small regularization term was added to the diagonal:

$$\Sigma_{reg} = \hat{\Sigma}_c + \epsilon I \quad (\text{where } \epsilon = 10^{-4})$$

1.3 Discriminant Analysis

Implemented discriminant functions $g_i(x)$ derived from the log-posterior.

Linear Discriminant Analysis (LDA): Assumed a common covariance matrix Σ (pooled from all classes). The decision boundary is linear.

$$g_i(x) = \mathbf{w}_i^T x + w_{i0}$$

Where

$$\mathbf{w}_i = \Sigma^{-1}\mu_i$$

Quadratic Discriminant Analysis (QDA): Used class-specific covariance matrices Σ_i. The decision boundary is quadratic.

$$g_i(x) = -\frac{1}{2}(x - \mu_i)^T\Sigma_i^{-1}(x - \mu_i) - \frac{1}{2}\ln|\Sigma_i| + \ln P(\omega_i)$$

2. Results

2.1 Classification Accuracy

The models were evaluated on the independent test set (300 samples total).

Model	Test Accuracy
LDA	88.33%
QDA	99.00%

Observation: QDA significantly outperforms LDA (99% vs 88.33%). This strongly suggests that the covariance structures (shapes and orientations) of the digits 0, 1, and 2 are distinct. LDA's assumption of a shared covariance matrix is too restrictive for this high-dimensional data, whereas QDA captures the specific variance of each digit class.

2.2 Discriminant Values for Sample Test Point

Analyzed the discriminant scores for the first sample in the test set.

True Class: 0

LDA Scores:

- Class 0: 262.80 (Highest)
- Class 1: -133.98
- Class 2: 153.92

Prediction: Class 0 (Correct)

QDA Scores:

- Class 0: -33,013.16 (Highest / Least Negative)
- Class 1: -315,782.29
- Class 2: -80,362.73

Prediction: Class 0 (Correct)

Note: The magnitude of QDA scores is large and negative because they represent log-probabilities in a 784-dimensional space, dominated by the term $-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)$.

3. Visualizations

3.1 t-SNE Plot: Training Set

The t-SNE visualization reduces the 784-dimensional data to 2 dimensions to visualize separability.

Analysis: The training data shows clear clustering for digits 0, 1, and 2. The clusters are well-separated, which explains why the Gaussian models (especially QDA) could fit the data well.

3.2 t-SNE Plot: Test Set

Analysis: The test set embeddings closely mirror the structure of the training set. The clear separation between Class 0 (Blue) and Class 1 (Orange) is evident. Class 2 (Green) is also distinct but shows slight proximity to Class 1 boundaries, which aligns with the discriminant values where Class 2 was the "second best" guess in the LDA example.

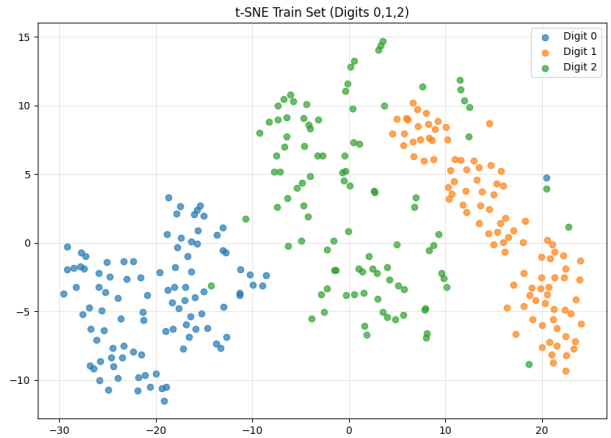

Figure 1: t-SNE Plot: Training Set

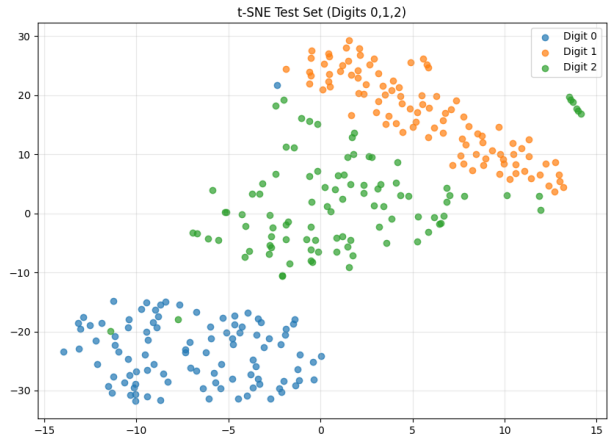

Figure 2: t-SNE Plot: Test Set